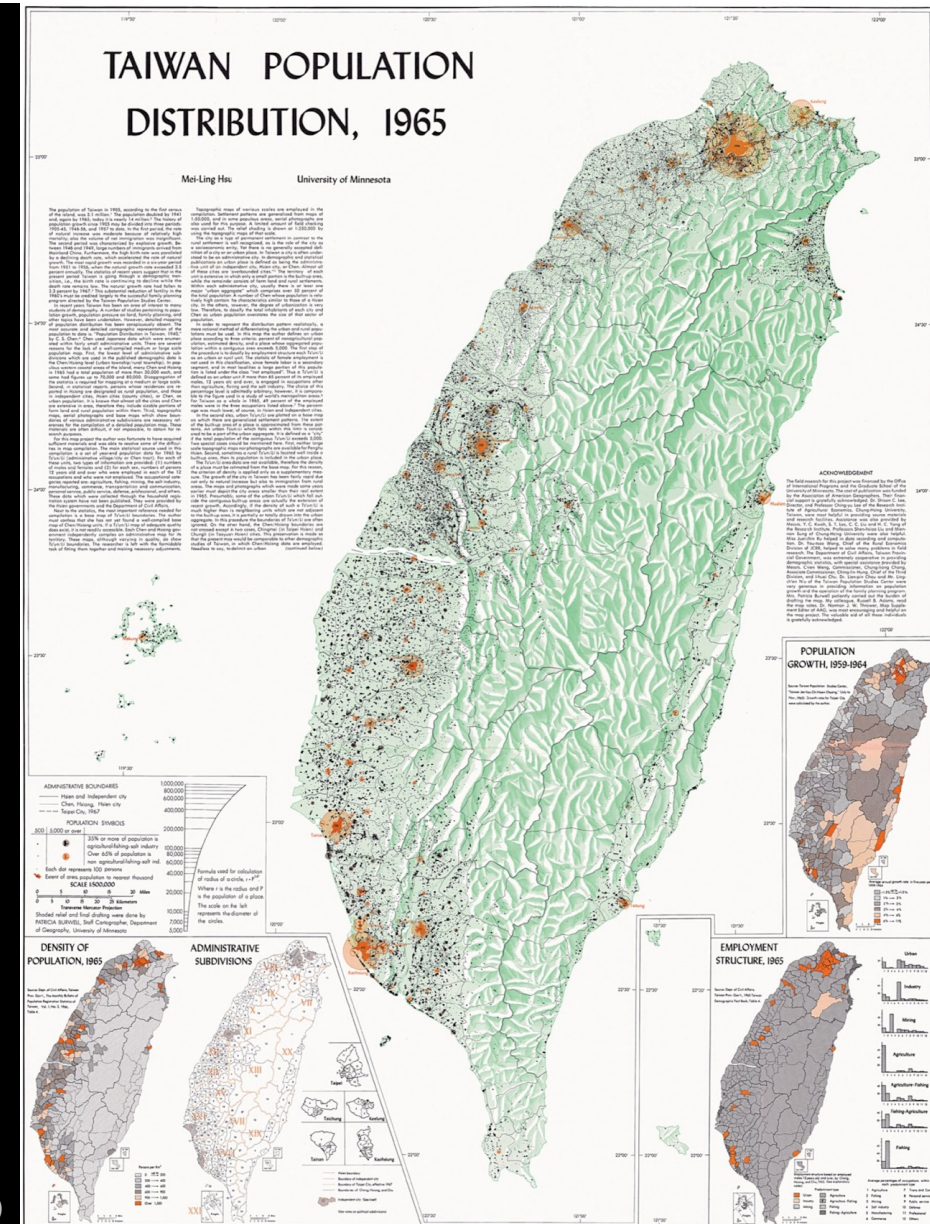


Attribute Data

GEOG 1026

Department of Geography
National Taiwan University

Hsu (1965)



Schedule

- April 17:
 - Quiz (coordinates and projection)
 - Project grouping
 - 2-hr Practice lab
- April 24: Midterm
 - 1-hr written exam
 - 2-hr Lab exam

期中考範圍

- March 6: Geographic Concept + Map History
 - March 13: Geographic Information + Map Composition
 - March 20: Coordinates + Projection
 - March 27: Geographic Representation + Thematic Maps
 - April 10: Attribute Data + Thematic Maps 2
-
- All Labs

期末地圖主題計畫書

- 組員人數：每組2-3人
- 繳交期限：May 8, 2:00 PM
- 繳交計畫書（pdf檔）與說明影片
- 計畫書構想：
 - 主題構想
 - 資料來源（如何蒐集資料）
 - 預計地圖呈現方式
- 計畫書說明影片：每組3分鐘
- 主題參考：
 - <https://wenlab501.github.io/Carto2016/>
 - <https://wenlab501.github.io/Carto2015/>

計畫書評分

- 同學們參與式評分，各組的計畫書（pdf檔）與影片提供檢視
- 評分標準：同學期待看到期末成果的程度（1-10分）
- 選出平均分數前50%的組別進行「期末口頭報告」
- 所有組別都需要繳交期末書面報告

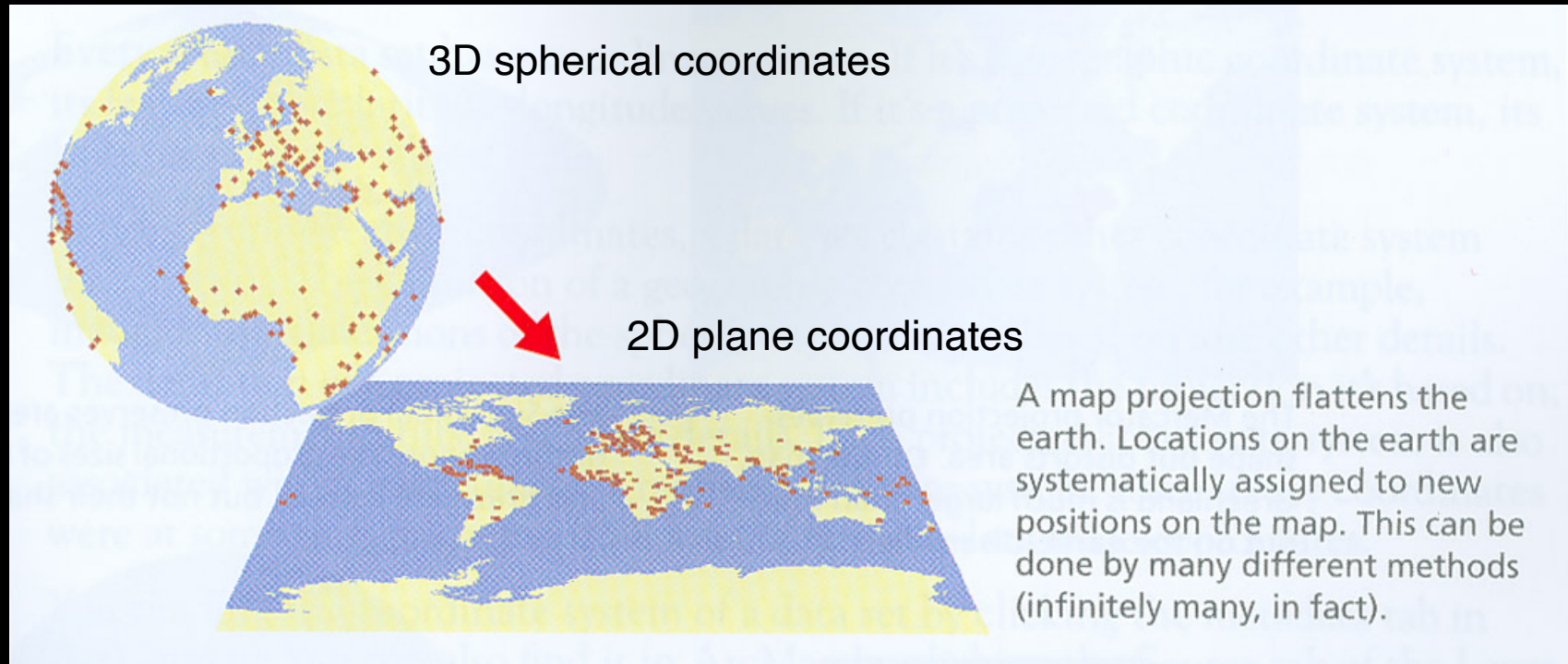
期末報告評分

- 期末競賽評分：所有同學評分 + TA評分
- 當週作業成績：
 - 入選組別的口頭報告
 - 未入選組別的評論報告

- 入選組別的口頭報告獎勵：
 - 一等：學期等第 A +
 - 二等：學期等第 (至少) A
 - 三等：期末考+50分
 - 其他入選組別：期末考+10分

- 未入選組別的書面報告獎勵：
 - 一等：學期等第 A +
 - 二等：學期等第 (至少) A
 - 三等：期末考+50分

Map Projection: There Is No Perfect Projection!



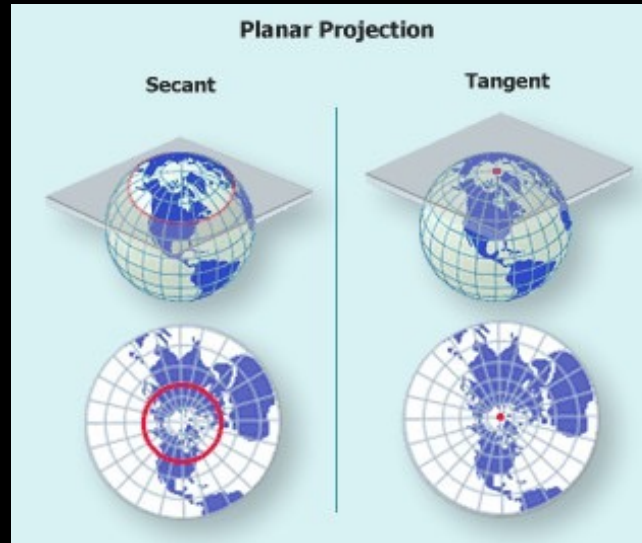
- Map projection is the orderly transfer of positions of places on the surface of the Earth to corresponding points on a flat sheet of paper, the map. (Frederick Pearson, 1990).

Key Concept

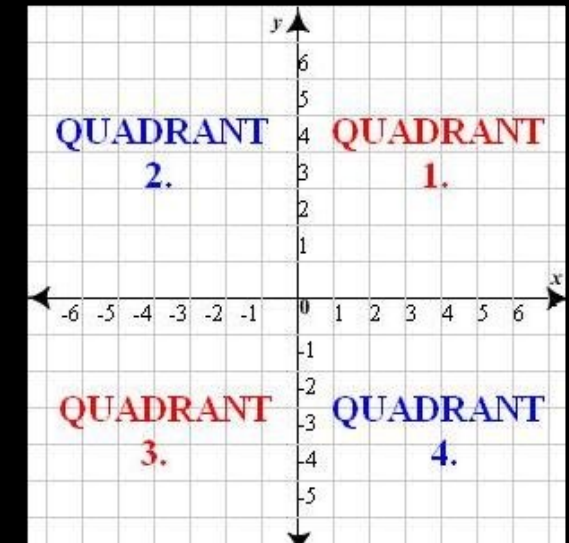


Geographic
Coordinate System

3D Spherical coordinates



Map
Projection



Plane
Coordinate

2D Plane coordinates

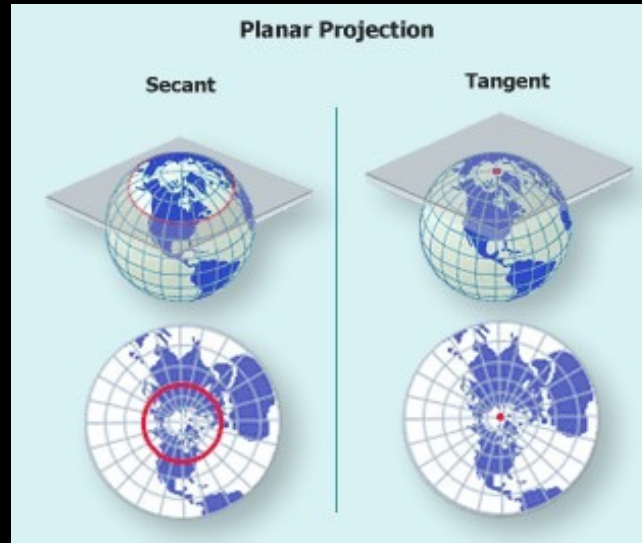
What should we consider when we convert 3D spherical coordinates to 2D plane coordinates?

Key Concept

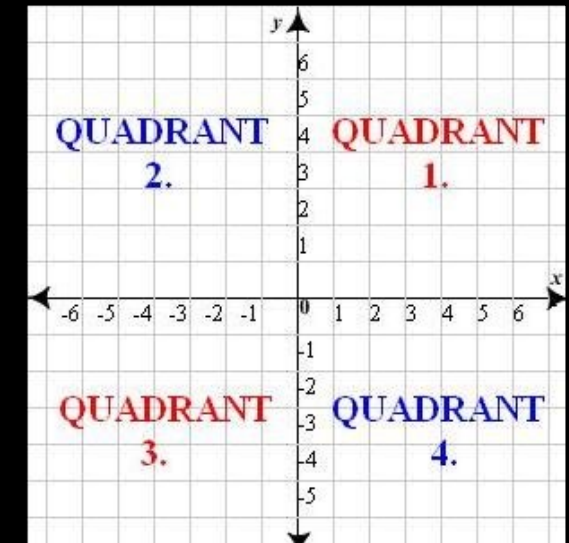


Geographic
Coordinate System

3D Spherical coordinates



Map
Projection



Plane
Coordinate

2D Plane coordinates

Key Concept



Geographic
Coordinate System

3D Spherical coordinates

Step 1: Define an Earth model first!

- Choose a shape
- Define the origin

Authalic sphere (標準球體)

Reference ellipsoid (參考橢球體)

Geoid (大地水準面)

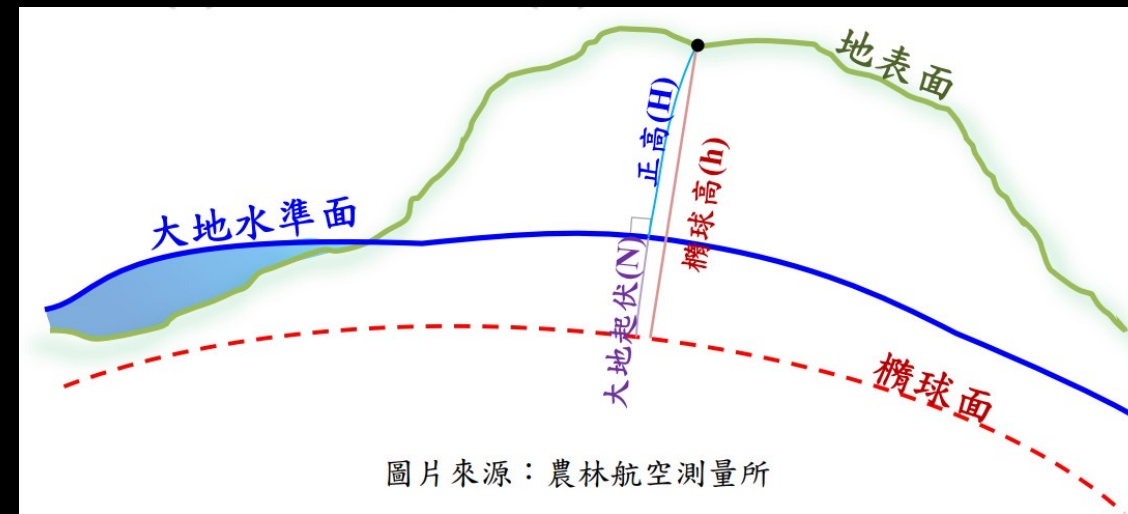
WGS84

GRS80

Clarke 1880

Bessel 1841

...



圖片來源：農林航空測量所

Key Concept

Step 2: Select a projection!

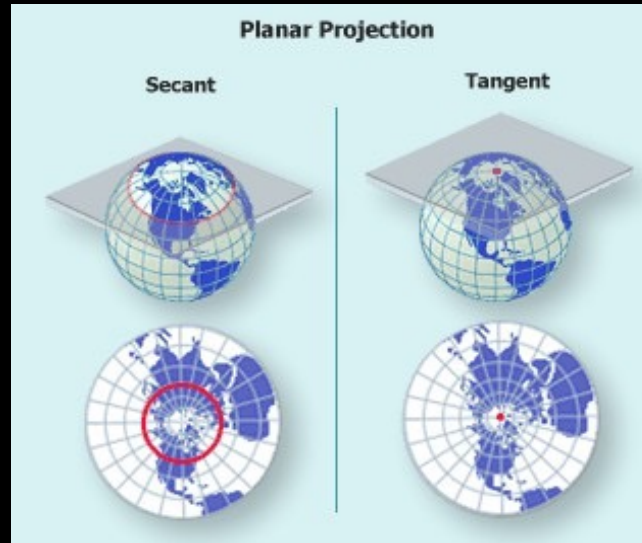


Geographic
Coordinate System

3D Spherical coordinates



Map
Projection



Purpose
Area
Spatial property
...

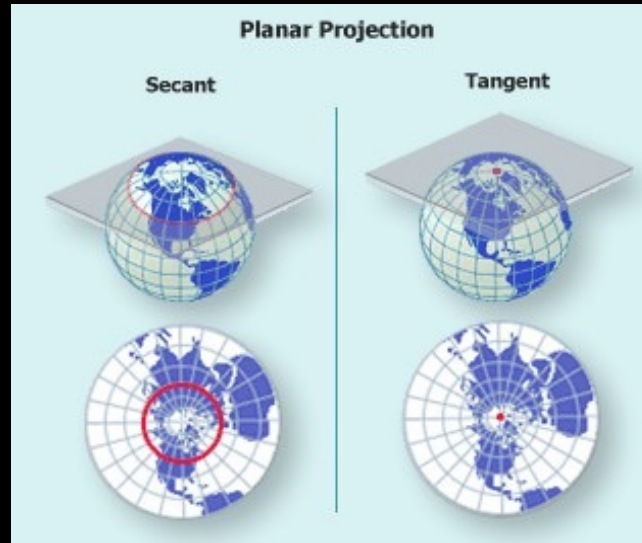
Key Concept

Step 3: Get the projected, plane coordinates



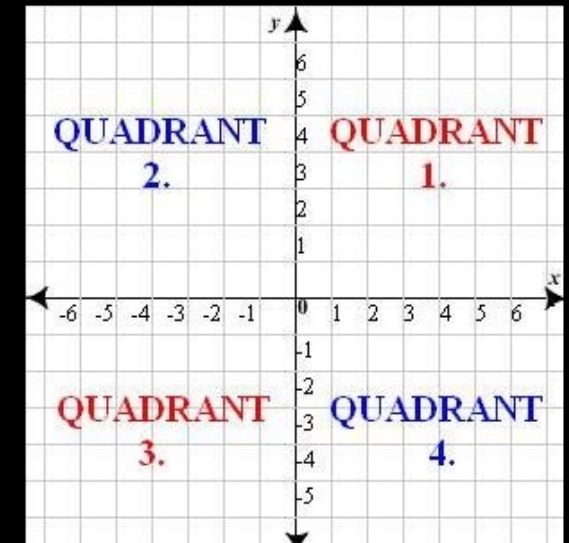
Geographic
Coordinate System

3D Spherical coordinates
Latitude & Longitude (經緯度)



Map
Projection

Ex. WGS84 + a projection



Plane
Coordinate

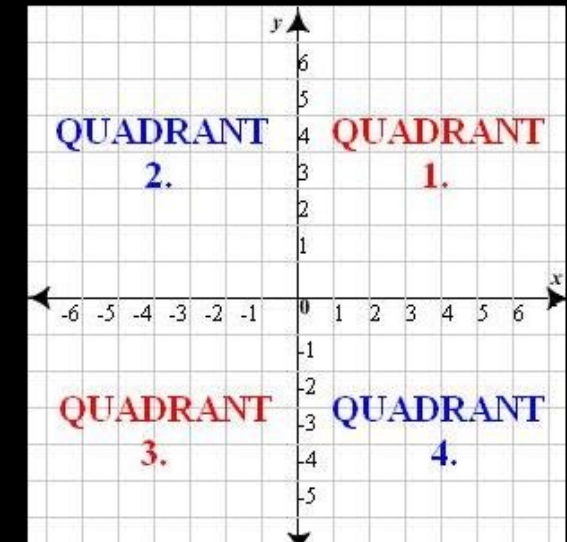
2D Plane coordinates
Latitude & Longitude (經緯度)

Key Concept

However, latitudes and longitudes are not suitable for calculating distance and area.

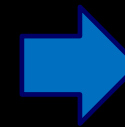
Alternative: Use Cartesian coordinates (unit: meter)

- | | | |
|-------------------------|---|---|
| Datum
(ex. TWD97) | { | • Step 1: |
| + | | • Define a reference ellipsoid (ex. GRS80) |
| Projection
(ex. TM2) | { | • Define an origin of the ellipsoid (ex. Earth's mass center) |
| | | • Step 2: |
| (ex. TMD97-TM2) | | • Select a projection (ex. TM2) |
| | | • Define a coordinate origin (ex. 0° , 121°E) |
| | | • Ex. False easting: 250,000 m; false northing: 0 m |



Coordinate
transformation
(坐標轉換)

Local Cartesian coordinate (直角坐標)

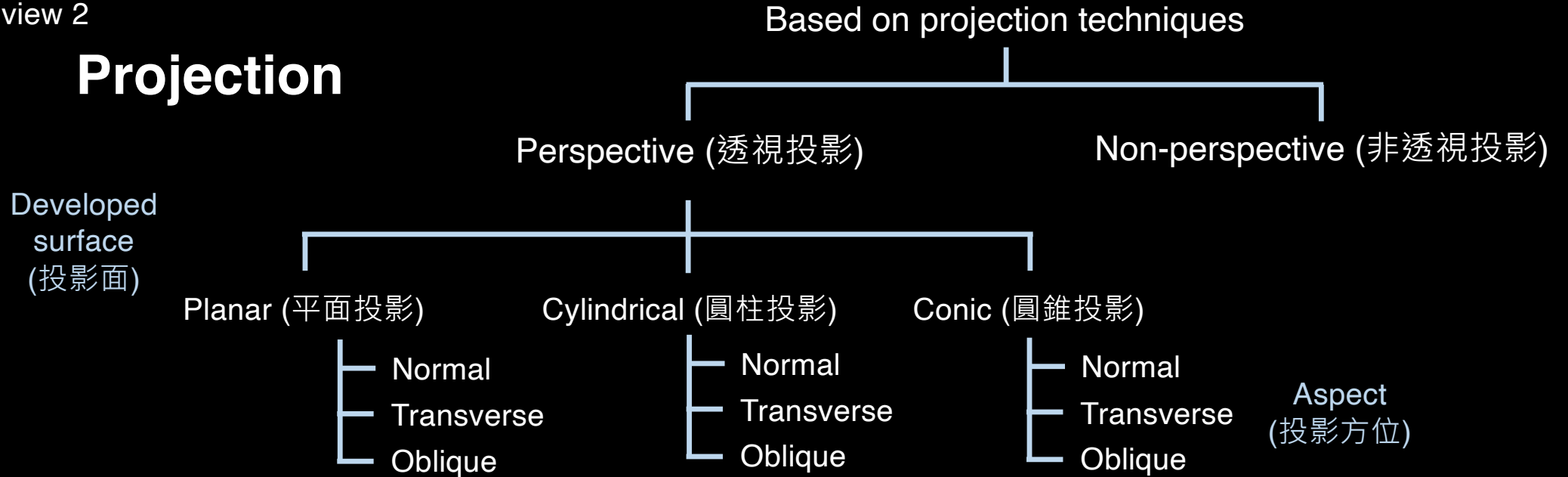


Plane
Coordinate

2D Plane coordinates

Latitude & Longitude (經緯度)

Projection

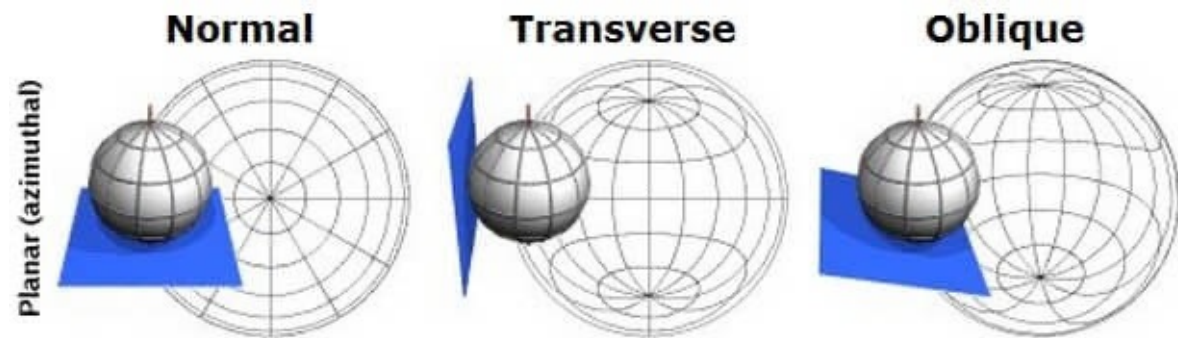


Aspect (refer to the developed surface)

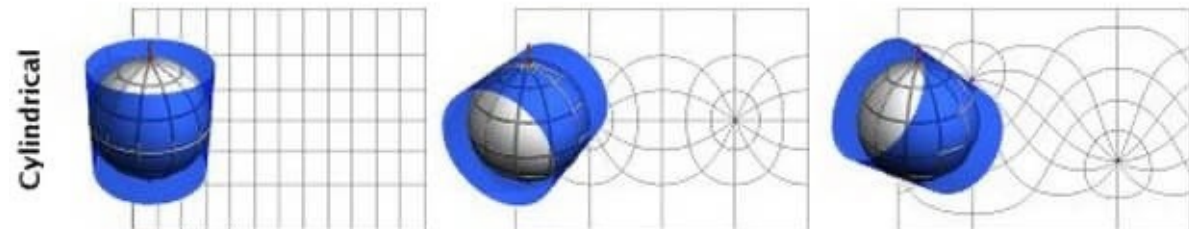
Developed surface

1. 平面 (方位投影)

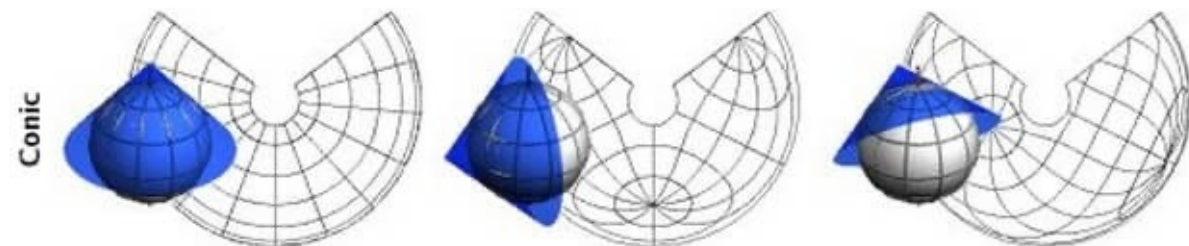
Light source variation



2. 圓柱投影



3. 圓錐投影



Projection

Based on projection techniques

Perspective (透視投影)

Non-perspective (非透視投影)

Developed
surface
(投影面)

Planar (平面投影)

Cylindrical (圓柱投影)

Conic (圓錐投影)

Azimuthal
(正方位投影)

Normal

Transverse

Oblique

Normal

Transverse

Oblique

Normal

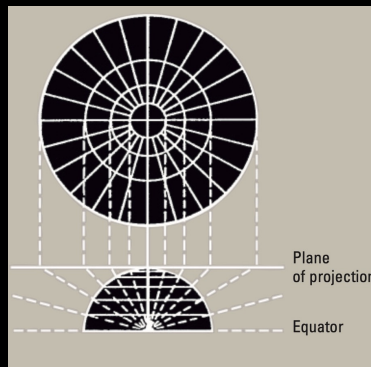
Transverse

Oblique

Aspect
(投影方位)

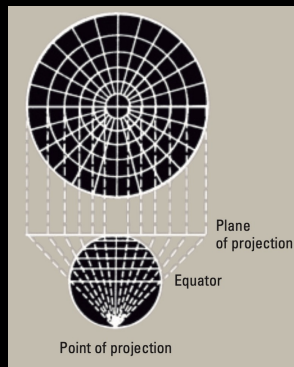
Light
source
(光源)

Gnomonic (心射投影)

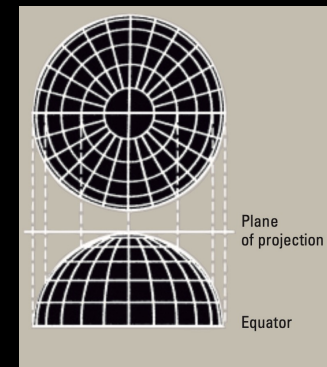


Stereographic (平射投影)

Conformal
(正形投影)



Orthographic (正射投影)



Projection

Based on spatial properties

Conformal or
orthomorphic
(正形投影)

Equal area or
equivalent
(等積投影)

Equidistant
(等距投影)

Azimuthal or
true-directional
(正方位投影)

Minimum error or
compromise
(折衷投影)

Example:
Stereographic (平射投影)
Mercator (麥卡托投影)
Transverse Mercator (橫麥卡托)
Lambert conformal conic

Example:
Hammer
Mollweide
Sinusoidal
Albers Equal-area conic

Example:
Equirectangular (方格投影)
Azimuthal equidistant
Equidistant conic

Example:
Gnomonic (心射投影)
Stereographic (平射投影)
Orthographic (正射投影)
Azimuthal equidistant

Example:
Robinson
Winkel Tripel
Miller cylindrical
Natural Earth

Geospatial Data Collection

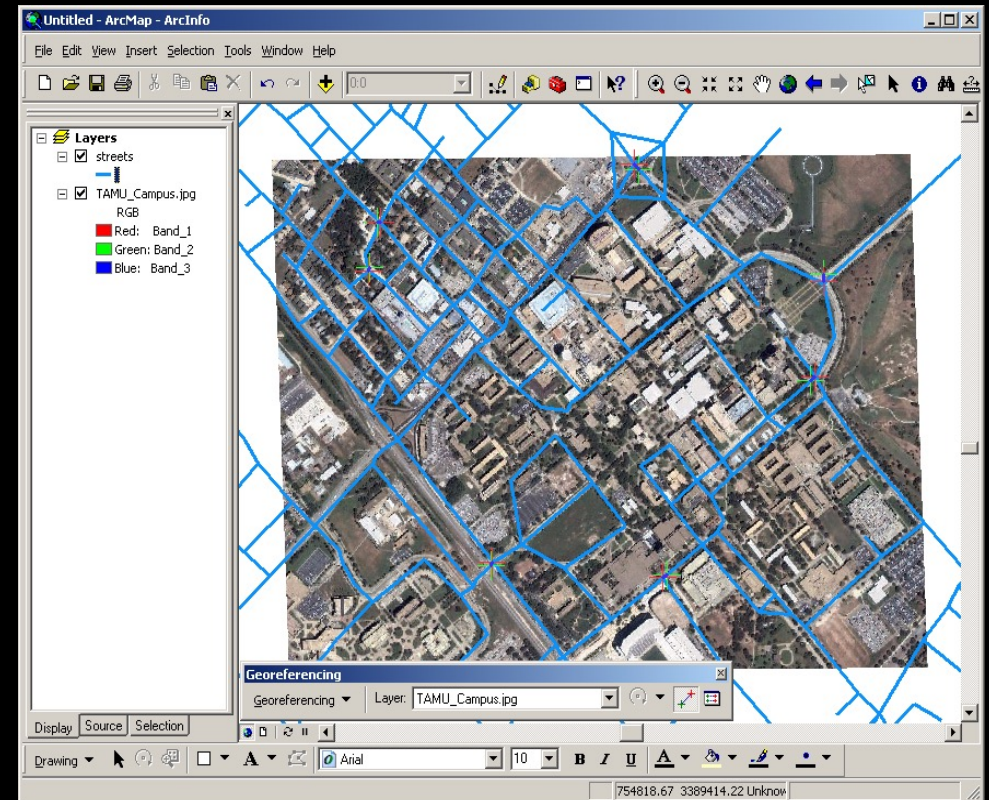
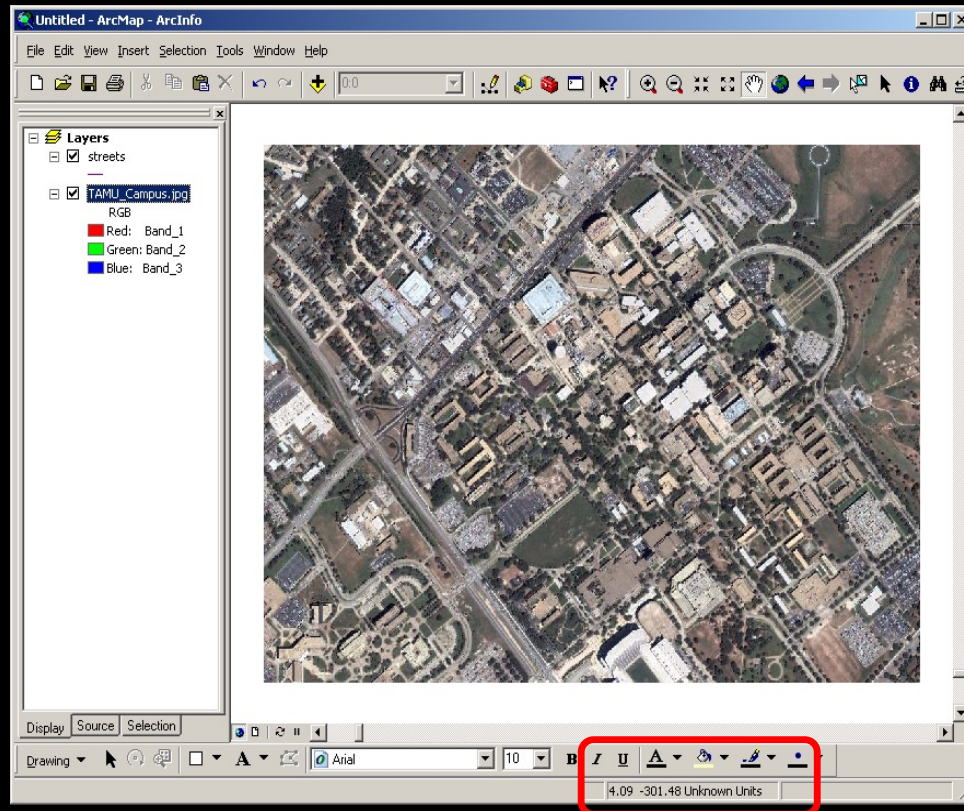
- **Primary data (direct measurements)**
 - Terrestrial geodesy
 - GPS/GNSS
 - (Geo-)physical observations
 - UAV
 - Web-mining
- **Secondary data**
 - Satellite images
 - Open data

Consideration of Data Integration

- Spatial property
 - Different coordinate systems → **Coordinate transformation!!!**
 - No coordinates → **Georeference!!!**
 - Different scale
 - Adjacent maps don't match
- Attribute property
 - Different time
 - Different attribute

Georeference (地理對位)

- Assigning coordinates to the page coordinates of a raster (image) or a planar map from a **known reference system**

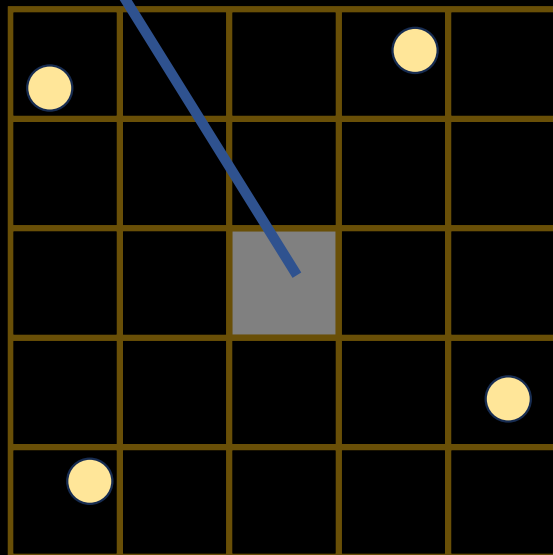


Rationale of Georeference

Image coordinates

Measured in rows and columns

(3, 3)

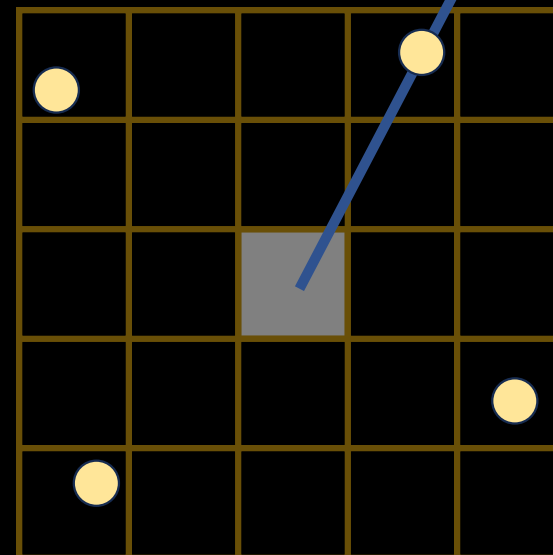


Ground control
point (地面控制點)

Map coordinates (WGS84)

Measured in lat/lon or easting/northing

(120.98, 23.97)



Transformation

Ground Control Point (GCP)

- Points on the Earth's surface where both coordinates can be identified.
- Use the measurements from geodesy (terrestrial, GNSS, etc.).
- Select GCPs from an **existing** geocoded images.
- The distribution of GCPs should be fairly **uniform** across your study area, covering as much of the image as possible, especially **edges and coners of the image**.

Last Week

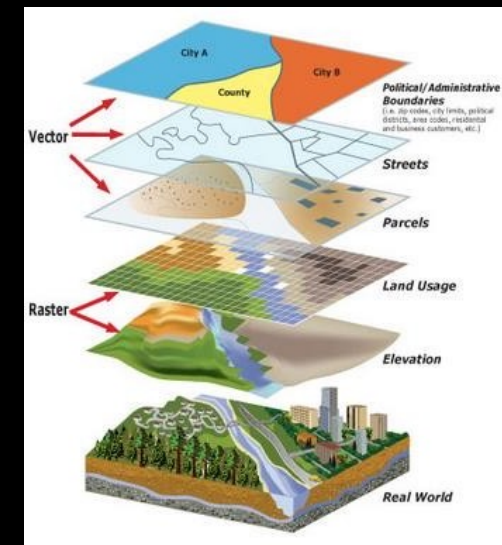
- The nature of geographic data
 - Spatial autocorrelation (空間自相關), distance decay (距離衰減), self-similarity (自我相似性), and scale (尺度)
 - Spatial sampling (空間取樣)
- Geographic Representation
 - Representation: Discrete objects (離散物件) and continuous fields (連續場)
 - Digital representation: raster data (網格資料) and vector data (向量資料)
 - Generalization (概括化)

This Week

- Attribute (屬性)
 - Nominal (類別), ordinal (序位或級序), interval (等距或間隔), ratio (比率)
- Query (查詢), join (連結), relate (關聯)
- Data transformation
 - Scale (尺度變換): aggregation (聚合)
 - Form (資料格式 / 型態轉換)
 - Boundary change (邊界變更): MAUP (可變區域單元問題) and BSU (最小統計區)

Category of Maps by Function

- General reference maps
 - General reference maps are to show the locations of a variety of different features, such as water bodies, coastlines, and roads.
- Thematic maps
 - Thematic maps or special purpose maps concentrates on **the distribution of a single attribute** or **the relationship among several**.
 - Basemap (底圖) + layers (圖層)
- Charts



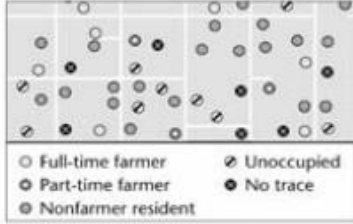
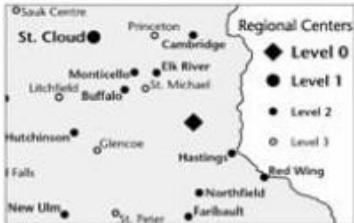
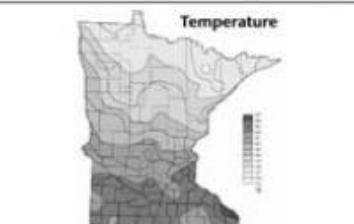
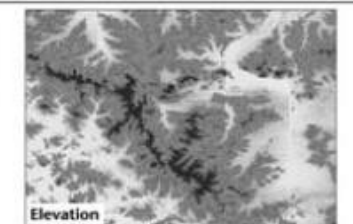
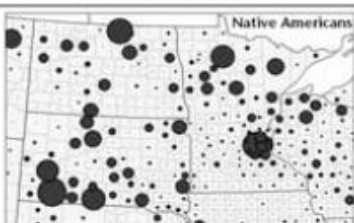
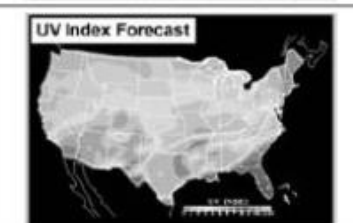
Attribute (屬性)

- Attributes generally refer to data used to describe, define, and explain geographic phenomena other than their spatial components.
- Map symbols often embody attribute information.
- Because attributes are non-graphic in nature, GIS commonly adopt database systems that are compatible with attribute data management.
- A table (表格) typically represents a collection of entities or relationships. Attributes are used to describe the properties of these entities or relationships.
- The collection of attributes contained in a table is referred to as a schema (綱要), which forms a unique structure.
- Each column (行 / 欄位) corresponds to a specific attribute, and each row (列) represents a particular instance (實體).
- Each attribute has its own data type, and values must be recorded according to the specified format.

Types of Attribute Data

- Nominal data (類別資料)
 - Only classification, and no order and quantitative magnitude
 - Gender, religion, land use, etc.
- Ordinal data (序位或級序資料)
 - Meaningful ranking or order, but no quantitative magnitude
 - Education level, etc.
- Interval data (等距或間隔資料)
 - Meaningful and equal difference between values, but no true zero
 - Temperature, elevation, etc.
- Ratio data (比率資料)
 - Meaningful ratio comparisons, and have a true zero
 - Income, population, etc.

Types of Attribute Data

	Description	Common Examples	Map Examples	
Nominal	qualitative measurements	name, type, state		
Ordinal	quantitative measurements with a clear order but without a defined zero value	small, medium, large		
Interval	quantitative measurements with a defined beginning point	temperature, height, distance		
Ratio	quantitative measurements that provide a relationship between two properties where the 0 value indicates the absence of the relationship	particulates mg/m ³ , time to cover a distance, dissolved oxygen in a liter of water, population density		

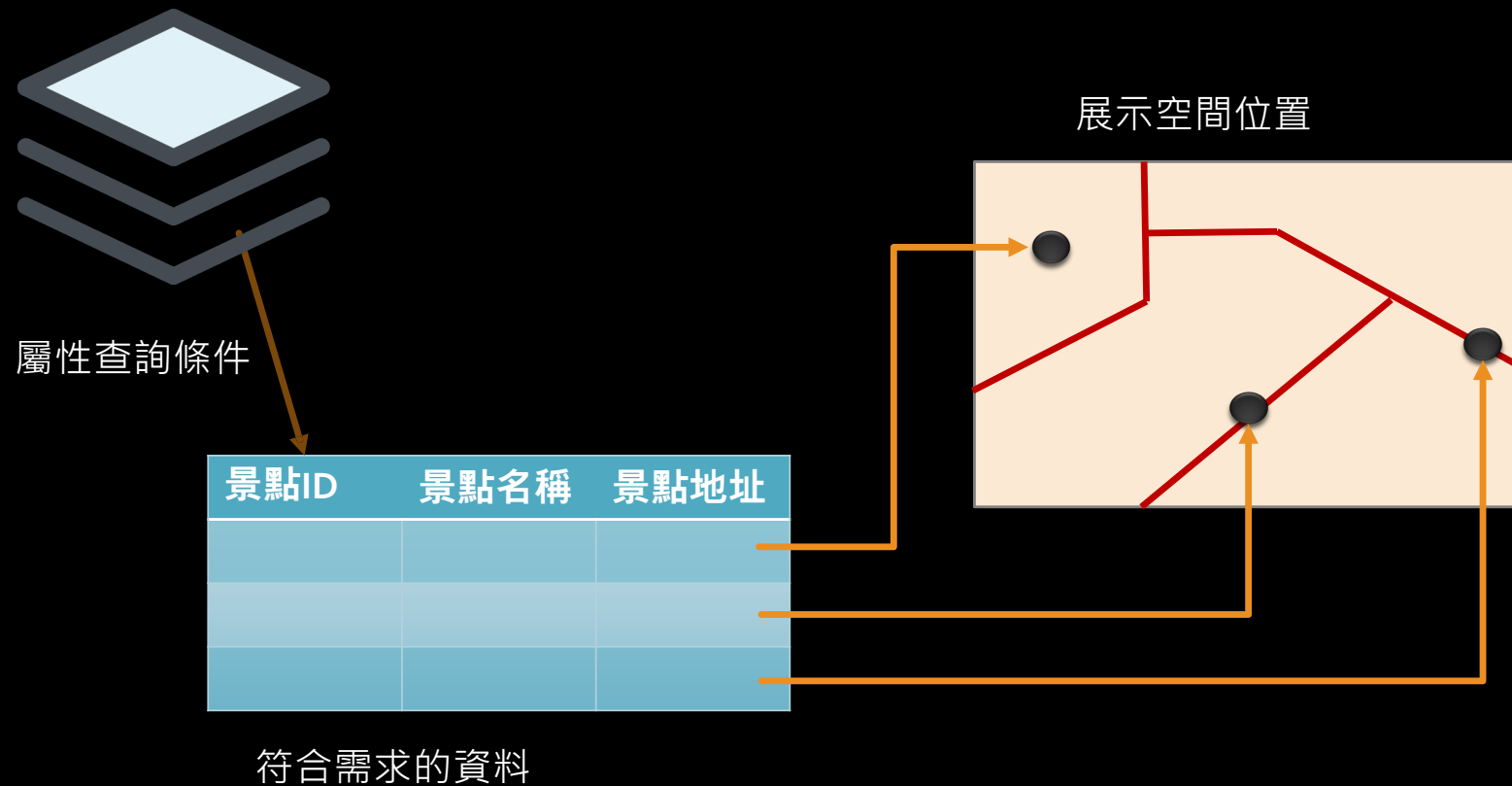
What can we do to the attribute data?

Query (查詢)

- Data query is the process of searching for and displaying specific data that meets certain criteria.
- The query itself acts as a constraint on the data been retrieved.
- Two main types of queries
 - Attribute query (non-spatial)
 - Filters data based on tabular information
 - Spatial query (geometric)
 - Filters data based on location or relationships, e.g. intersect, touch (share a boundary), buffer, inside a boundary, etc.
- Although spatial and attribute data are conceptually distinct, they must be integrated and treated as representations of real-world entities, and a linkage between the two must be established.
- Once the linkage is established, attribute-based queries can be used to derive spatial distribution of features, and vice versa.

Query (查詢)

- Example: attribute-based queries to derive spatial distribution of features



Integration of Spatial and Attribute Data

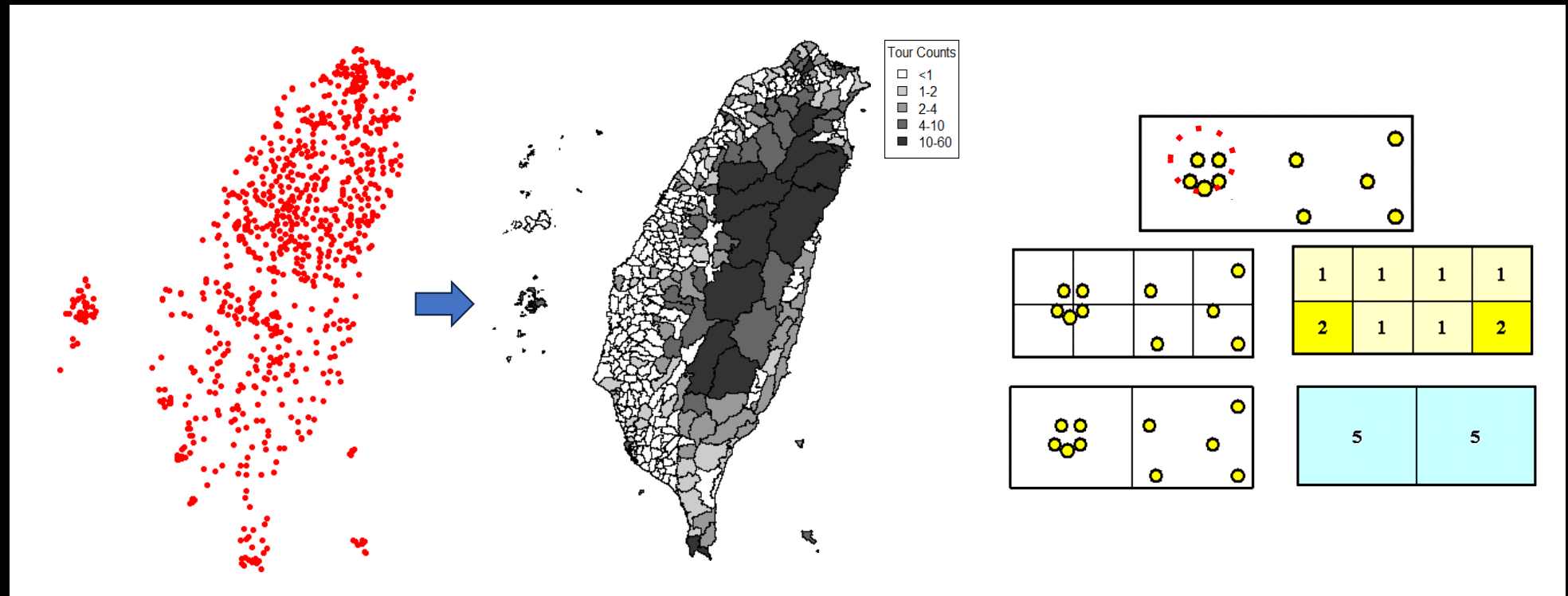
- Join (連結)
 - Merge two tables through “keys” (鍵)
 - Key: ID, county names, etc.
- Relate (關聯)
 - The two tables remain physically separate, but when you select a record in Table A, the corresponding data in Table B is automatically selected.

Data Transformation (資料轉換) in GIS

- Scale (尺度變換)
- Form (資料格式 / 型態轉換)
- Boundary change (邊界變更)

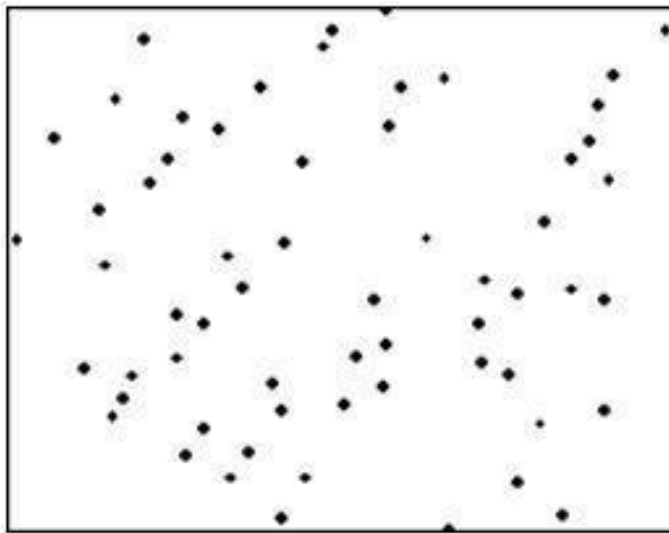
Scale (尺度變換)

- Aggregation (聚合)
 - From smaller units into larger ones. For example, aggregating the population of individual households into the administrative district level.

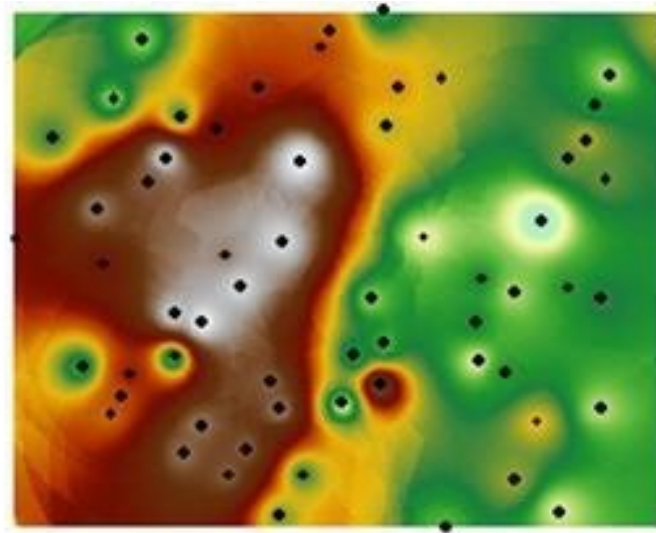


Form (資料格式 / 型態轉換)

- Vector to raster, raster to vector, etc.
- Spatial interpolation (空間內插): point to surface



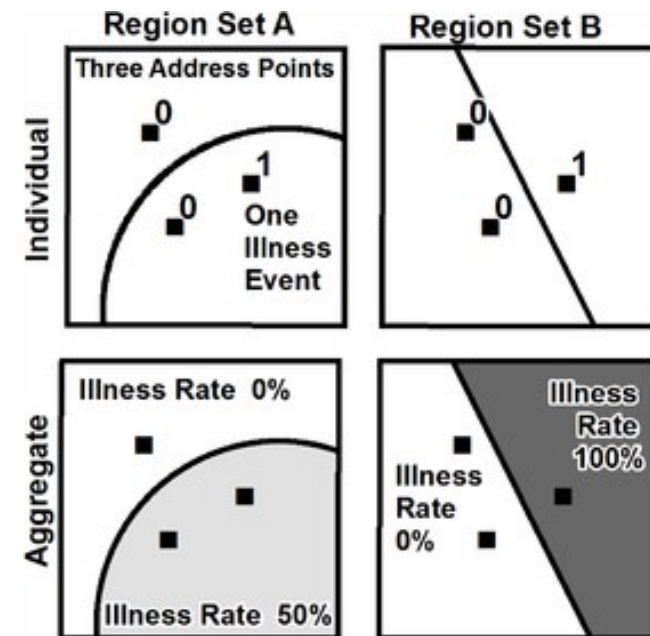
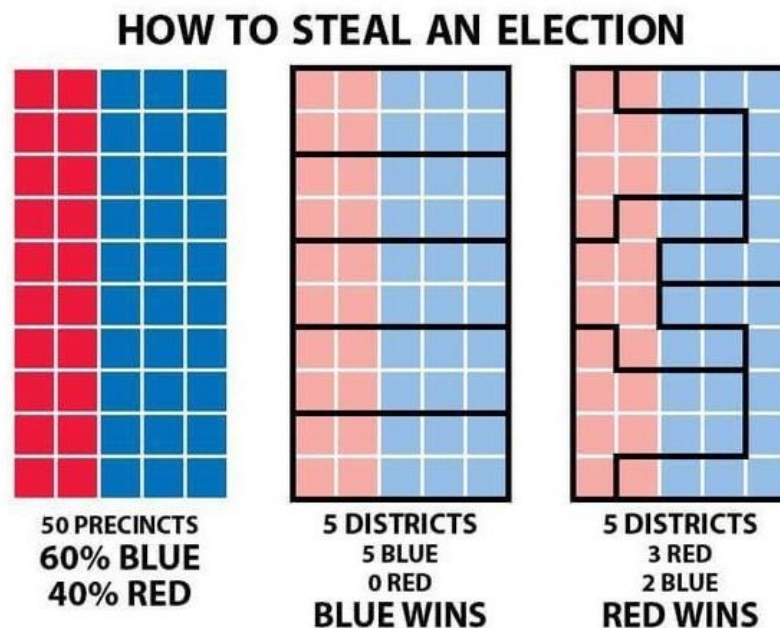
the sample points



the output data

Boundary Change (邊界變更)

- Modifiable areal unit problem (MAUP) (可變區域單元問題)
 - The same basic data yield different results when aggregated in different ways



Basic Statistical Unit (BSU) (最小統計區)

- The smallest geographical area used for collecting and disseminating census or statistical data.
- It serves as the “building block” for larger units. Multiple BSUs are aggregated to form administrative districts (like village or townships).
- The primary purpose of a BSU is to provide high-resolution spatial data while protecting individual privacy (data masking).
- Statistics are only released if the BSU meets a minimum population or household threshold (閾值) to prevent the identification of specific persons.
- BSUs have stable boundaries over time and do not overlap.

內政部統計區分類系統

**社會經濟資料服務平台**
· 國土資訊系統 ·

會員登入

關於SEGIS ▾ 推動成果 ▾ 資料與服務 ▾ 統計地圖 ▾

現在位置：首頁 > 關於SEGIS > 統計區分類系統

統計區分類系統

[沿革與發展](#)[階層架構](#)[作業標準](#)[資料說明](#)[圖層下載](#)[統計區比對服務](#)[應用案例圖集](#)

沿革與發展

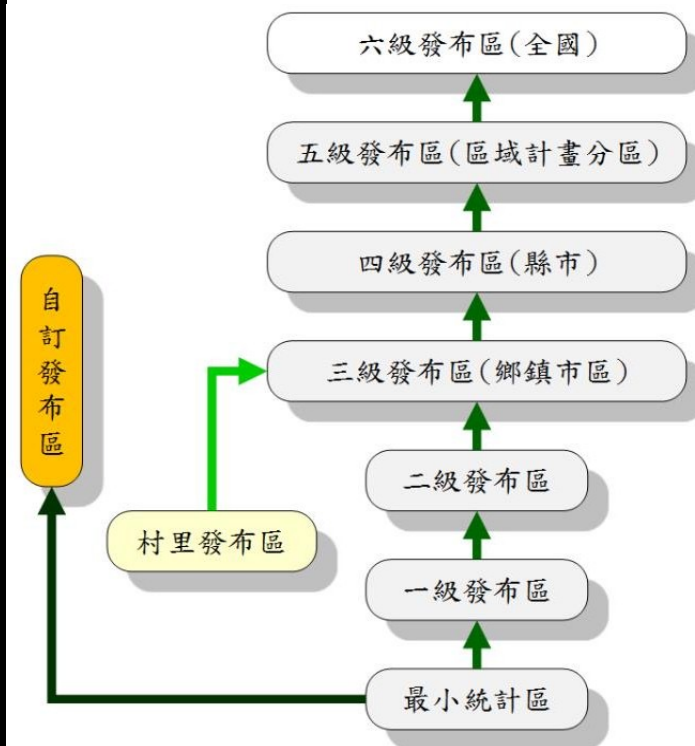
以往我國社會經濟資料，均以行政區來進行統計資料之蒐集、整理、分析；而行政區之設計是以便利行政管理為原則，經過整合之行政區域統計資料，無法再進行精緻小區域或特殊地域統計分析；再者，行政區會視行政管理需要進行界線調整、合併或重行劃分等情況，導致資料單元在時間序列上不一致性，降低資料之時間序列分析效果。

建置長期固定性小型區域或特定區域作為統計專用之最小空間單元，以保障個案資料隱私。進而整合賦予各項社會經濟資料各階層統計區之空間特性，以利強化與深化社經資料之空間分析，提升統計資料內涵與應用潛力，在先進國家已有陸續辦理達30多年歷史。

「國土資訊系統統計區建置計畫」係前行政院經濟建設委員會「國土資訊系統推動小組」所訂「國家地理資訊系統建置及推動十年計畫」執行計畫之一，分由內政部統計處負責推動辦理，計畫在國土資訊系統基礎環境數化成熟後，利用街道巷弄、門牌地址或經緯度等對位功能及區劃效果，建置通用性長久固定之最小統計區基礎圖資，俾利整合社會經濟各項公務登記及調查統計資料，轉化為具有空間分布之統計資訊，再利用各階層統計區地理圖資萃取社經資料分布及變遷趨勢資訊，提升社經統計資料之應用，供為政府及民間單位各項政策規劃與決策重要參據。

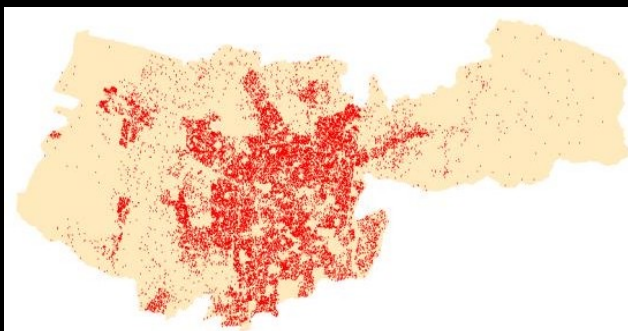
https://segis.moi.gov.tw/STAT/Web/Portal/GroupProfile/STAT_GroupProfileSTATClass.aspx

- 六級發布區：全國
- 五級發布區：區域計畫分區
- 四級發布區：縣市層級
- 三級發布區：鄉鎮市區層級
- 二級發布區：每一區域理想人口數為3000人
- 一級發布區：每一區域理想人口數約450人
- 最小統計區：每一最小統計區人數約小於450人

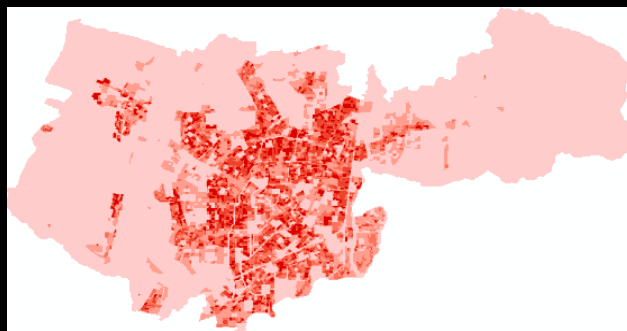


範例：比較各種尺度下的人口分布

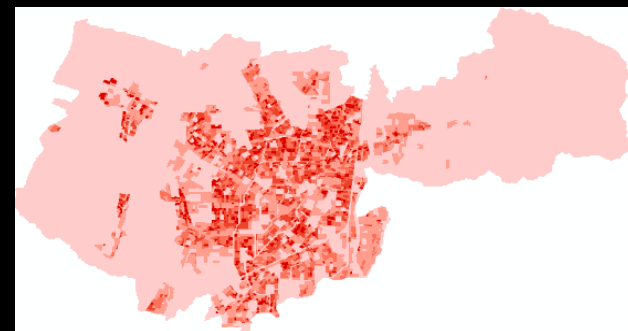
點資料



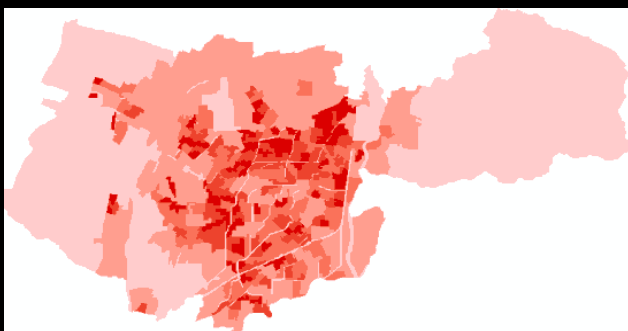
最小統計區



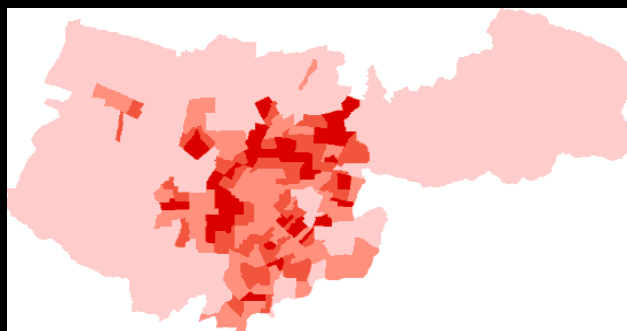
一級發布區



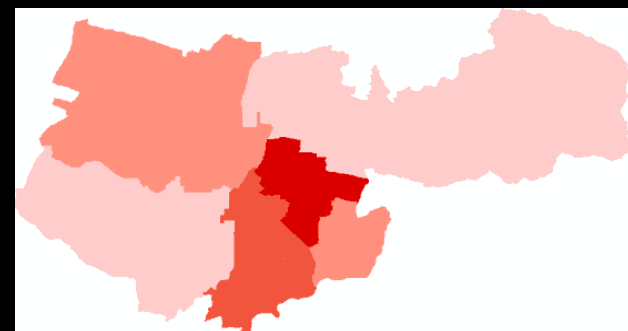
二級發布區



村里統計



鄉鎮市區統計



Basic GIS Analysis

- Overlay analysis (疊圖分析)
- Network analysis (路徑分析)
- Buffer analysis (環域分析)
- Terrain analysis (地形 / 地勢分析)

